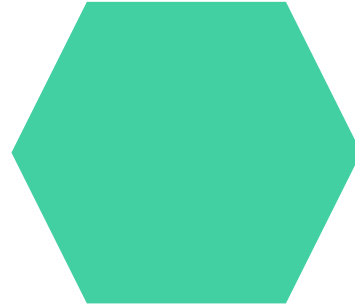
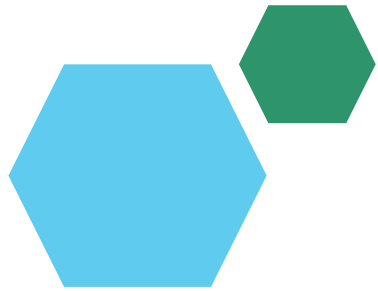




Project Title - Healthcare Analytics For Doctor Visit TIRTC

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College: RNS Institute Of Technology



PROBLEM STATEMENT

Healthcare providers need to understand the factors associated with doctor visits to identify patterns in healthcare utilization. This project analyzes patient demographic, health, illness, chronic-condition and insurance-related information to understand how these factors are associated with the number of doctor visits.



Project Description

The Healthcare Analytics for Doctor Visits project focuses on analyzing patterns in healthcare utilization using a dataset containing 5,190 patient records.

The dataset includes information related to doctor visits, gender, age, income, illness, reduced activity due to illness, health condition, insurance coverage and chronic conditions.

The project involves data cleaning, exploratory data analysis, descriptive statistics, grouping and data visualization using Python. Different patient and health-related factors are compared with the number of doctor visits to identify meaningful patterns and relationships.

The analysis helps understand which health-related factors are associated with higher levels of healthcare utilization and provides data-driven insights that can support healthcare planning and decision-making.

WHO ARE THE END USERS?

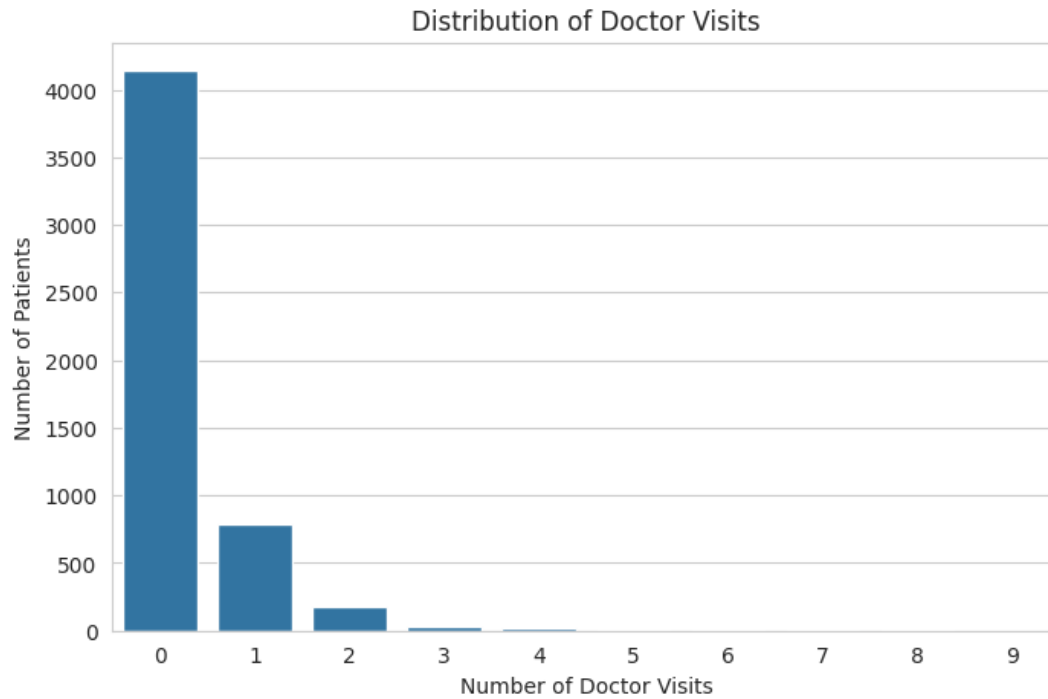
- **Hospitals & Healthcare Providers**
Understand patient visit patterns and healthcare utilization.
- **Healthcare Administrators**
Support resource allocation and service planning.
- **Health Insurance Organizations**
Understand healthcare utilization patterns across patient groups.
- **Public Health Organizations**
Support healthcare planning and population-level analysis.
- **Healthcare Analysts & Researchers**
Study relationships between health conditions and healthcare utilization.
- **Policy & Planning Teams**
Use data-driven insights for healthcare planning.

Technology Used

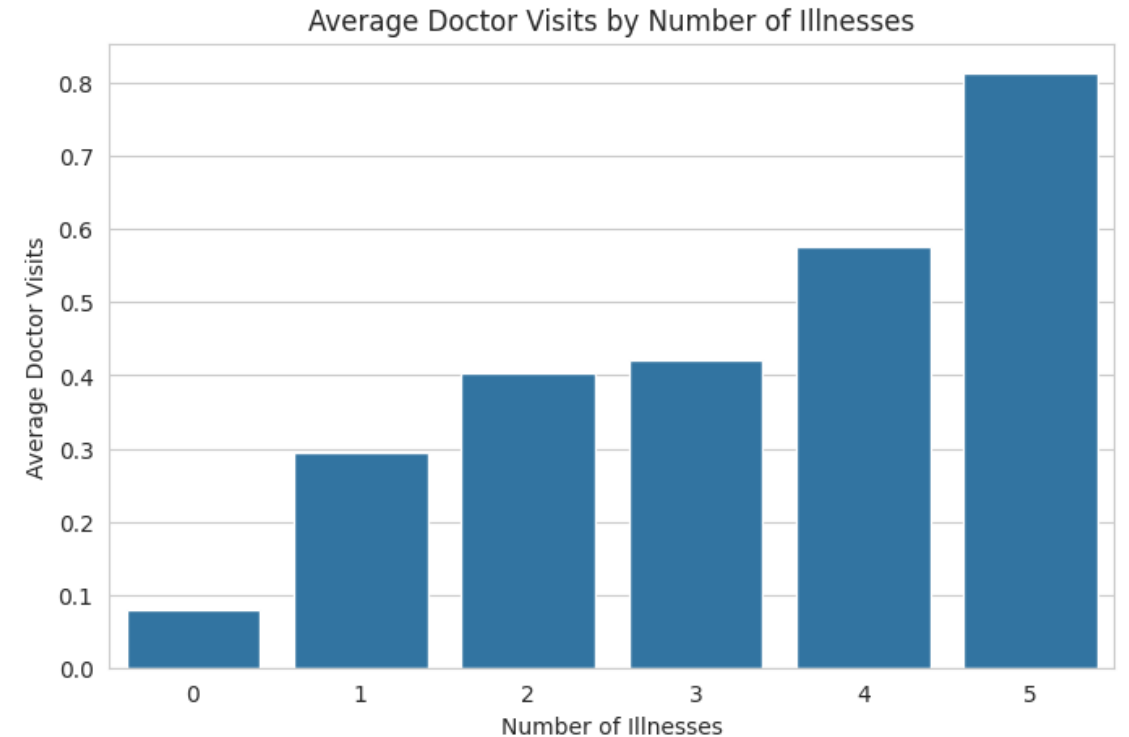
- **Python**
Data analysis and processing
- **Pandas**
Data cleaning, manipulation and aggregation
- **NumPy**
Numerical operations
- **Matplotlib**
Data visualization
- **Seaborn**
Statistical visualization and charts
- **Google Colab**
Cloud-based environment for developing and executing the Python analysis



RESULTS: Doctor Visit Patterns

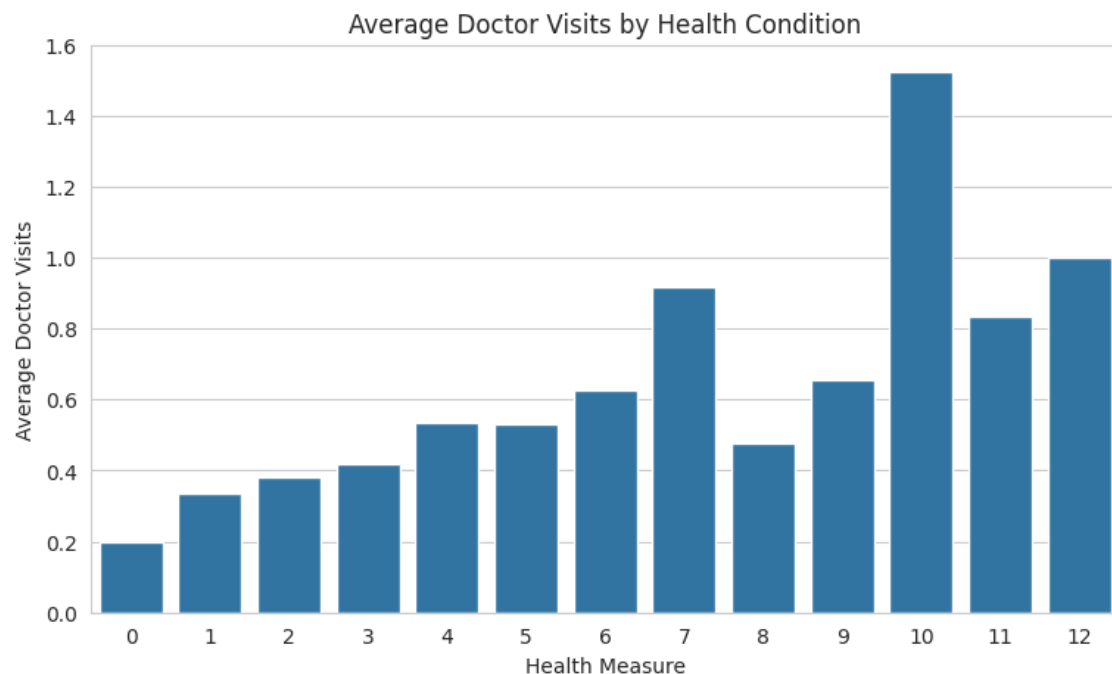


Key Insight: Most respondents reported zero doctor visits, while higher visit counts were considerably less frequent.



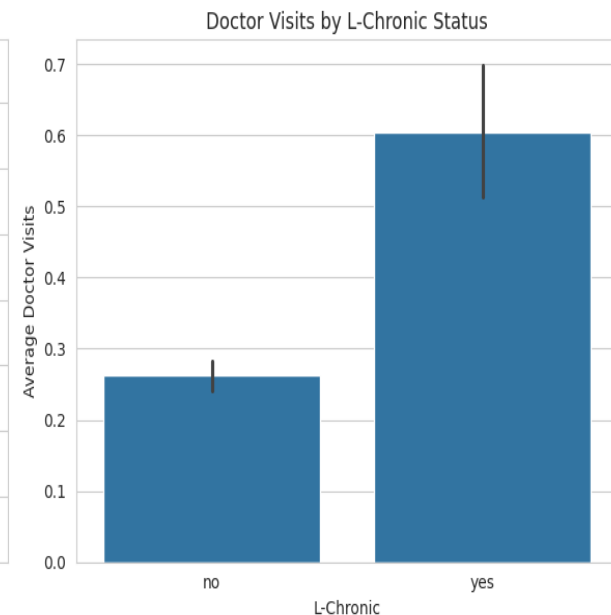
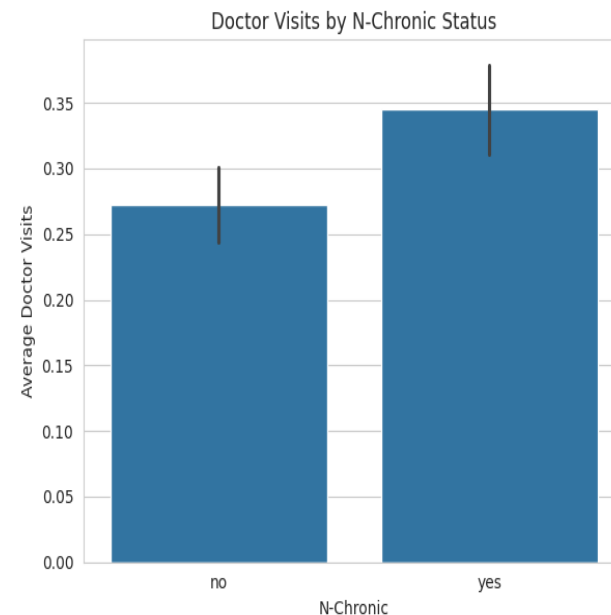
Key Insight: Average doctor visits generally increased as the number of reported illnesses increased.

RESULTS: Health & Chronic Conditions



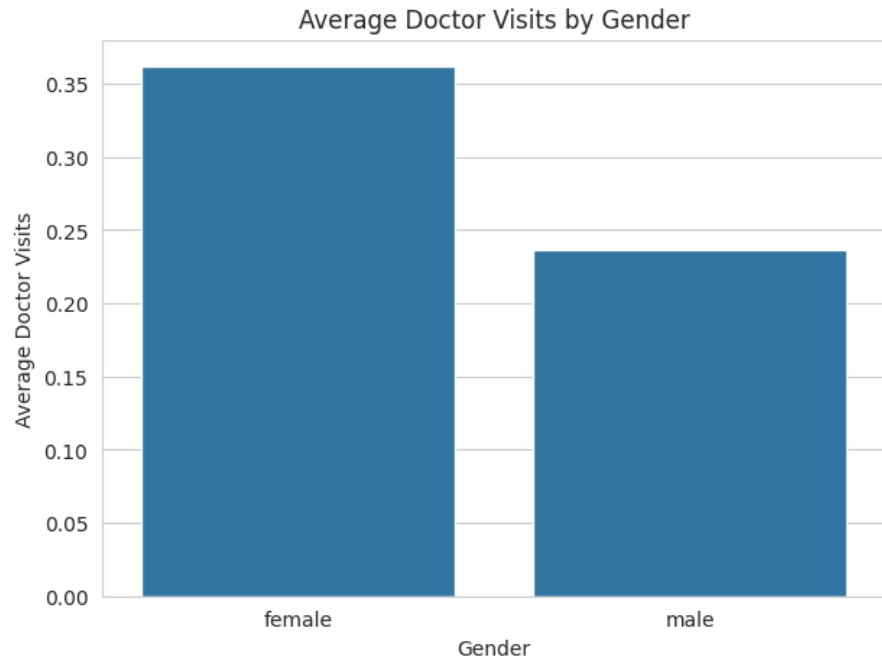
Key Insight: Doctor visits generally showed an upward pattern across higher recorded health-condition levels.

Overall Observation: Health-related factors showed stronger associations with doctor visits than demographic or income-related variables.

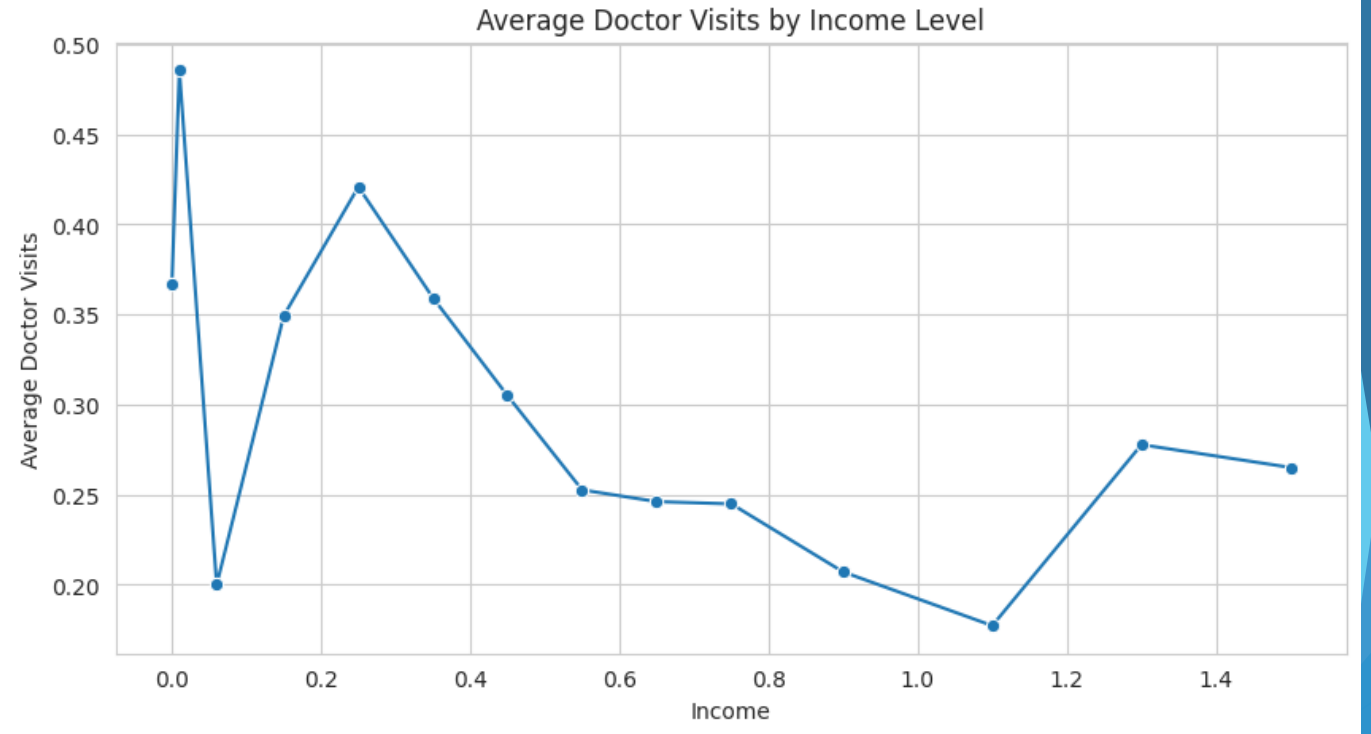


Key Insight: Respondents with the recorded L-chronic condition had a higher average number of doctor visits than those without it.

RESULTS

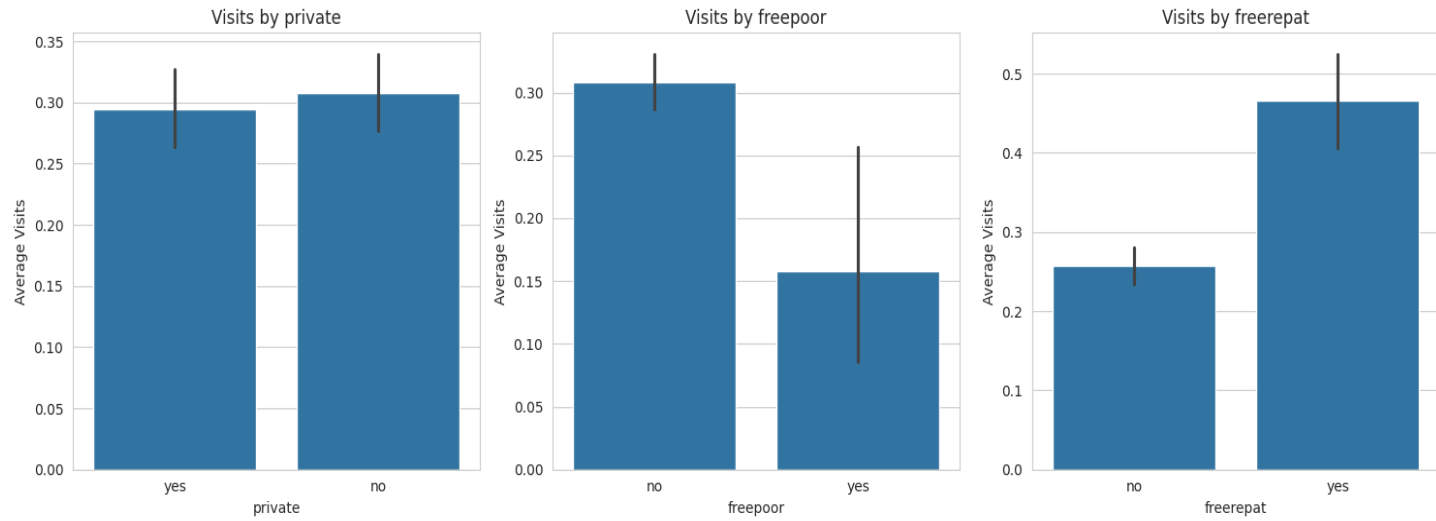


Female respondents recorded a higher average number of doctor visits than male respondents.

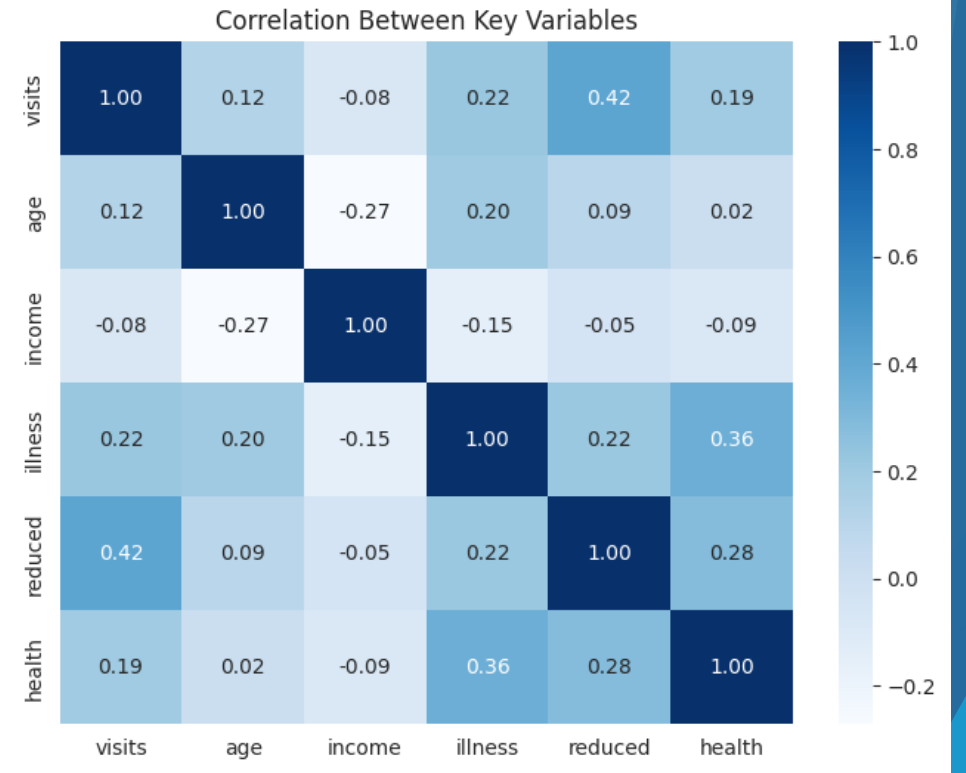


Average doctor visits varied across income levels, but the relationship was not consistently upward or downward.

RESULTS



Doctor visit patterns varied across the recorded insurance and healthcare-access categories.



Illness, reduced activity and health-related variables showed stronger associations with doctor visits than income.

Thank you

Healthcare Analytics for Doctor Visits

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